

Jaspreet Singh Nahal

Delhi, India | jaspreetnahal100@gmail.com | +91-9999604054

linkedin.com/in/jaspreet-nahal | github.com/jaspreet-nahal | jaspreetnahal.xyz | leetcode.com/jaspreet-nahal

SUMMARY

AI/ML Engineer with a published arXiv paper on a 205M-parameter GPT Transformer pretrained from scratch in PyTorch. Delivered production RAG and data-visualization systems serving 10,000+ documents, and currently building Aria, a local-first autonomous agent architecture with a React-based visualization frontend using LangGraph and FastAPI.

EXPERIENCE

Research Intern, Agentic AI & LLMs

Aug 2026 – Nov 2026 (Expected)

IIT Kharagpur, Dept. of CSE

- Developing Jantram, an LLM-agent-based fact-checking pipeline in LangGraph, implementing a multi-agent workflow (claim extraction, evidence retrieval, verification) with conditional graph edges for insert/discard/flag decisions
- Proposed hash-based duplicate detection for claim IDs, redesigned evidence storage to preserve traceable source URLs with confidence scores, and evaluated semantic-similarity claim matching versus keyword-based matching to improve conflict detection

AI & Cloud Engineer Intern

Jul 2026 – Sep 2026

Cloud Inn Tech

Remote

- Architecting the backend for an AI Tender & RFP Copilot on a three-tier data architecture (PostgreSQL, ChromaDB, AWS S3) supporting RAG-based document intelligence for government tenders
- Designing a pluggable LLM provider layer (Ollama for local development, Gemini for production) behind a rule-based eligibility engine built with the compliance team, keeping decisions deterministic and auditable
- Implementing multi-tenant isolation via PostgreSQL Row-Level Security and vector-store metadata filtering, with retry-with-backoff and circuit-breaker resilience patterns

Machine Learning Intern

Aug 2025 – Oct 2025

Expedien

Noida, India

- Engineered a RAG-based legal query resolver with the engineering team using LangChain, FAISS, and Hugging Face Transformers over a corpus of 10,000+ legal documents
- Optimized the retrieval pipeline with vector embeddings and semantic search, achieving sub-2s response time and 40% higher throughput, and applied fine-tuning to reduce hallucinations

PROJECTS

EverydayGPT | [arXiv:2606.11212](https://arxiv.org/abs/2606.11212) | [Live Demo](#) | [GitHub](#) | *PyTorch, FAISS, RAG, NLP*

Feb 2026

- Built a routing system that decides between fast retrieval and full GPT generation, skipping generation for 85% of queries and cutting average latency 6.3× while maintaining answer quality
- Pretrained a 205M-parameter GPT from scratch on a curated subset of a 10B-token corpus, benchmarked against 8 baselines including LangChain RAG; trained model runs inference on consumer CPU under 2 GB of RAM

CodeAtlas | [Live Demo](#) | [GitHub](#) | *Python, LangChain, RAG, NLP, NetworkX*

Mar 2026

- Developed an AI-powered code intelligence platform with interactive dependency graph visualization over GitHub repos and ZIP archives, enabling natural language Q&A and codebase mapping across 8 languages
- Added domain-aware file classification and an adaptive personalization layer that tailors code explanations to user proficiency using retrieval-augmented context

Aria | *LangGraph, FastAPI, Ollama, RAG, FAISS, React, Python* | *In Progress*

Jun 2026

- Building a local-first autonomous desktop agent (React frontend) that converts natural language into OS-level task execution across 3 domains – files, email, and browser – with zero cloud dependency
- Architecting a multi-agent debate layer via LangGraph with a tiered safety model (auto-execute/confirm/reject) and persistent memory via FastAPI, Ollama, and FAISS

EDUCATION

Bachelor of Technology in Computer Science Engineering (AI)

2023 – 2027

Dr. A.P.J. Abdul Kalam Technical University (AKTU, formerly Uttar Pradesh Technical University)

Uttar Pradesh, India

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript

Machine Learning & AI: PyTorch, Hugging Face Transformers, SentenceTransformers, FAISS, LangChain, LangGraph, RAG, NLP, Instruction Fine-Tuning, TensorFlow

Data Science: NumPy, Pandas, scikit-learn, Matplotlib, Hugging Face Datasets

Visualization & Frontend: React, NetworkX, Streamlit

Backend, Cloud & DevOps: FastAPI, Flask, REST APIs, PostgreSQL, MongoDB, ChromaDB, AWS (S3, EC2/IAM), Docker, Git, CI/CD

PUBLICATIONS

Nahal, J. S. (2026). **EverydayGPT: Confidence-Gated Routing for Efficient and Safe Hybrid GPT–RAG Conversational QA**. arXiv:2606.11212 [cs.CL] — arxiv.org/abs/2606.11212

ACHIEVEMENTS & CERTIFICATIONS

Awards: 1st, Code Farming Codeathon (750+ participants); 3rd, HackIndia Spark-4 Voyager Blitz (900+ participants, 2026)

Hackathons — IBM AI Hackathon, Google Solutions Challenge, GUVI Hackathon (2025)

Certifications: Google – Prompt Design in Vertex AI, Responsible AI Practices; IBM – Accelerating Deep Learning with GPUs; BCG X – GenAI Financial Chatbot